MGMTMSA-400-01/02: Homework 1 Solution

Due Date: October 11, 2026, 11:59 PM

Question 1: Law of Large Numbers

(a) Generate one sample for each sample size $N = 10$, 100, 1000, and 10000. For each sample, report the sample mean $\bar{Y}$ and its absolute difference from the population mean, $|\bar{Y} - \mu|$, in a table.

```r
set.seed(123)
mu <- 0

# Generate one sample for each sample size and compute its mean.
mean_10 <- mean(runif(10, min = -5, max = 5))
mean_100 <- mean(runif(100, min = -5, max = 5))
mean_1000 <- mean(runif(1000, min = -5, max = 5))
mean_10000 <- mean(runif(10000, min = -5, max = 5))

# Put the results into a table.
results_a <- data.frame(
  N = c(10, 100, 1000, 10000),
  Sample_mean = c(mean_10, mean_100, mean_1000, mean_10000)
)
results_a$Absolute_difference <- abs(results_a$Sample_mean - mu)

round(results_a, 3)
```

```
##       N Sample_mean Absolute_difference
## 1    10       0.782               0.782
## 2   100      -0.010               0.010
## 3  1000      -0.035               0.035
## 4 10000      -0.018               0.018
```

Table 1: Sample Means and Absolute Differences

Sample Size N	Sample Mean $\bar{Y}$	Absolute Difference $\lvert\bar{Y} - \mu\rvert$
10	0.782	0.782
100	-0.010	0.010
1000	-0.035	0.035
10000	-0.018	0.018

(b) As sample size N increases, would you generally expect the absolute difference $|\bar{Y} - \mu|$ to become smaller or larger? Which theorem explains this expectation? Discuss using the

results from (a).

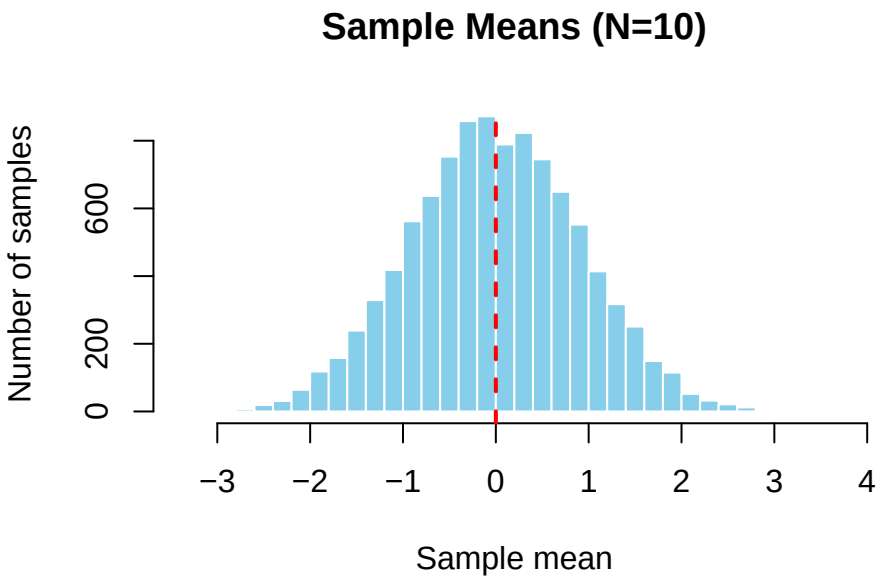
With larger sample size, the sample mean generally becomes closer to the population mean of zero. This is the **Law of Large Numbers**. In our results, the absolute difference falls from 0.782 at $N = 10$ to 0.018 at $N = 10000$.

(Optional) It does not decrease at every step: the difference increases between $N = 100$ and $N = 1000$. This small increase is caused by the sampling error: the $N = 1000$ sample may happen to include a few data points that are very distant from the population mean of zero.

(c) Generate 10000 samples, each with $N = 10$ observations, and compute the mean of each sample. Plot a histogram of the resulting 10000 sample means.

```r
# Repeat 10000 times: draw 10 observations and compute their mean.
sample_means_10 <- replicate(10000, mean(runif(10, min=-5, max=5)))

hist(sample_means_10, breaks=30, col="skyblue", border="white",
     main="Sample Means (N=10)", xlab="Sample mean",
     ylab="Number of samples")
abline(v = mu, col = "red", lwd = 2, lty = 2)
```



(d) Which theorem explains why the sample means in (c) have an approximately normal distribution, even though the original observations are uniformly distributed?

The sample means have an approximately normal distribution, even though the observations are drawn from a uniform distribution. This illustrates the **Central Limit Theorem**: with a sufficiently large sample size, the sample mean has an approximately normal distribution centered around the population mean. In this uniform distribution example, the histogram already looks approximately normal with $N = 10$.

(e) Repeat the simulation in (c) using sample sizes $N = 100$ and $N = 1000$, generating 10000 samples for each sample size. For each of the three sample sizes ($N = 10$, 100, and 1000), compute the 25th and 90th percentiles of the 10000 sample means. Report your results in a table. You may reuse your results from (c) for $N = 10$.

In R, you can compute both percentiles using `quantile(sample_means, c(0.25, 0.90))`.

```r
# Generate sample means for the two larger sample sizes.
sample_means_100 <- replicate(10000, mean(runif(100, min=-5, max=5)))
sample_means_1000 <- replicate(10000, mean(runif(1000, min=-5, max=5)))

# Reuse the N = 10 sample means from part (c).
percentiles_10 <- quantile(sample_means_10, c(0.25, 0.90))
percentiles_100 <- quantile(sample_means_100, c(0.25, 0.90))
percentiles_1000 <- quantile(sample_means_1000, c(0.25, 0.90))

# Put the percentiles into a table.
results_e <- data.frame(
  N = c(10, 100, 1000),
  Percentile_25 = c(
    percentiles_10[1], percentiles_100[1], percentiles_1000[1]
  ),
  Percentile_90 = c(
    percentiles_10[2], percentiles_100[2], percentiles_1000[2]
  )
)

round(results_e, 3)
```

```
##      N Percentile_25 Percentile_90
## 1   10        -0.626         1.176
## 2  100        -0.195         0.360
## 3 1000        -0.064         0.117
```

Table 2: Percentiles of Sample Means under Different Sample Sizes

Sample Size N	25th Percentile	90th Percentile
10	-0.626	1.176
100	-0.195	0.360
1000	-0.064	0.117

(f) As N increases, do the 25th and 90th percentiles in (e) move closer to or farther from the population mean of zero? Explain intuitively why increasing N leads to this pattern. (3 points)

Both percentiles move closer to zero as N increases. In a small sample, a few unusually high or low observations can have a large influence on the sample mean. In a larger sample, each observation has less influence, and values above and below zero tend to balance out. In other words, the sample mean becomes a more precise estimate of the population mean, with smaller sampling errors in general. As a result, the sample means are less spread out and more concentrated around zero. The 25th percentile therefore becomes less negative, while the 90th percentile becomes smaller, bringing both closer to zero.

Question 2: Jane's All-You-Can-Eat Restaurant

(a) What is the best point predictor of the cost for the next customer? (2 points)

The best point predictor is the population mean, **$17.50**.

(b) Jane wants to find a 90% prediction interval of customer costs, centered around the mean ($17.50). In other words, 90% of customers' costs should fall within this range. What is the width of that interval, i.e., the distance between the upper and lower bounds? (2 points)

```
price <- 19
mean_cost <- 17.50
sd_cost <- 3

# A central 90% interval leaves 5% in each tail.
z <- qnorm(0.95)
lower_bound <- mean_cost - z * sd_cost
upper_bound <- mean_cost + z * sd_cost
interval_width <- upper_bound - lower_bound
cat("90% prediction interval: [$", round(lower_bound, 2),
    ", $", round(upper_bound, 2), "]\n", sep="")
```

```
## 90% prediction interval: [$12.57, $22.43]
```

```
cat("Interval width: $", round(interval_width, 2), "\n", sep="")
```

```
## Interval width: $9.87
```

The central 90% prediction interval is $17.50 \pm 1.645 \times 3$, or approximately **[$12.57, $22.43]**. Its width is $2 \times 1.645 \times 3 \approx$ **$9.87**. We use the standard deviation of individual customer costs, $3.00, because we are predicting one customer's cost rather than estimating the population mean.

(c) Suppose customers are i.i.d. On one day, Jane serves 100 customers, and on another, she serves 250 customers. Which day is more likely to have a higher average profit **per customer** (price minus cost per customer)? (2 points)

```
# Subtracting cost from a fixed price changes the mean, but not the SD.
mean_profit <- price - mean_cost
sd_profit <- sd_cost
# Standard deviation of average profit for each day.
sd_average_100 <- sd_profit / sqrt(100)
sd_average_250 <- sd_profit / sqrt(250)
cat("Expected average profit per customer: $", mean_profit, "\n", sep="")
```

```
## Expected average profit per customer: $1.5
```

```
cat("SD of average profit, N=100:", round(sd_average_100, 4), "\n")
```

```
## SD of average profit, N=100: 0.3
```

```
cat("SD of average profit, N=250:", round(sd_average_250, 4), "\n")
```

```
## SD of average profit, N=250: 0.1897
```

Neither day is more likely to have a higher average profit per customer. Both days have an expected average profit of $19 - 17.50 =$ $1.50. Their average profits are normally distributed, and the difference between the two days' average profits is normally distributed around zero. Therefore, each day has a **50% chance** of having the higher average profit. Serving 250 customers makes average profit more concentrated around $1.50; it does not make average profit systematically higher.

(d) Suppose customers are i.i.d. On one day, Jane serves 100 customers, and on another, she serves 250 customers. Which day is more likely to have an average profit **per customer** (price minus cost per customer) that is higher than $2.00 per customer? (2 points)

```
# lower.tail=FALSE gives the probability of being above the cutoff.
probability_100 <- pnorm(2, mean=mean_profit, sd=sd_average_100, lower.tail=FALSE)
probability_250 <- pnorm(2, mean=mean_profit, sd=sd_average_250, lower.tail=FALSE)
cat("Probability above $2.00, N=100:", round(100 * probability_100, 2), "%\n")
```

```
## Probability above $2.00, N=100: 4.78 %
```

```
cat("Probability above $2.00, N=250:", round(100 * probability_250, 2), "%\n")
```

```
## Probability above $2.00, N=250: 0.42 %
```

The day with **100 customers** is more likely to have average profit above $2.00 per customer. Its average profit is more spread out around $1.50, so it has a greater chance of exceeding a cutoff above $1.50. With 250 customers, average profit stays closer to $1.50. The probabilities are approximately **4.78%** for 100 customers and **0.42%** for 250 customers.

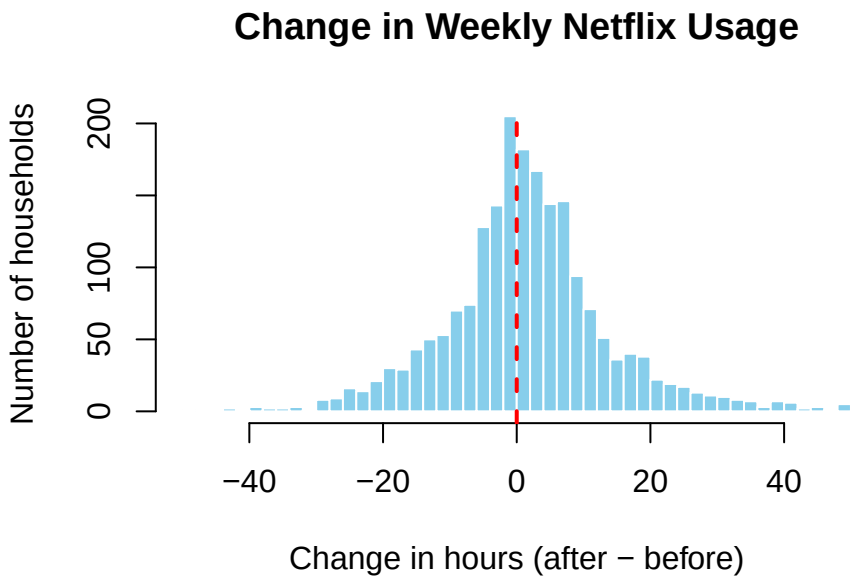
You may also answer this question without the computations above. Notice that $2.00 is higher than the expected mean profit of $1.50 per customer. For average profit to exceed $2.00, the sample mean must exceed the population mean by more than $0.50. Such a positive sampling error is more likely with a smaller sample because the sample mean is more variable. Therefore, the probability of average profit exceeding $2.00 per customer is higher when Jane serves 100 customers.

Question 3: Netflix Game Launch

(a) Changes in Netflix Usage Among Subscribers

(a-1) For each household in this group, calculate the change in Netflix usage by subtracting `netflix_hours_pre` from `netflix_hours_post`. Plot a histogram of these changes and briefly describe its shape. (2 points)

```
streaming_beta <- read.csv("streaming_beta.csv")
# Keep only households subscribed both before AND after the launch.
subscribers <- streaming_beta[
  streaming_beta$has_netflix_pre == 1 &
  streaming_beta$has_netflix_post == 1,
]
hours_change <- subscribers$netflix_hours_post -
                subscribers$netflix_hours_pre
hist(hours_change, breaks=40, col="skyblue", border="white",
     main="Change in Weekly Netflix Usage",
     xlab="Change in hours (after - before)",
     ylab="Number of households")
abline(v=0, col="red", lwd=2, lty=2)
```




```
cat("Number of households:", length(hours_change), "\n")
```

```
## Number of households: 2012
```

The histogram shows both positive and negative changes. Most changes are near zero, with fewer households experiencing large increases or decreases. Positive values indicate increased usage, while negative values indicate decreased usage. The analysis includes 2012 households that subscribed in both periods.

(a-2) Should you use a one-tailed or two-tailed hypothesis test? State the null and alternative hypotheses, and explain what the population mean in your hypotheses represents. Remember that the data are measured in hours. (2 points)

Use a **one-tailed test**, because we want to know whether the mean increase is **greater than** 60 minutes. Let μ be the population mean change in weekly Netflix usage per household among households subscribed both before and after the launch, measured in hours. Since 60 minutes equals 1 hours, the hypotheses are:

$$H_0 : \mu \leq 1 \qquad \text{versus} \qquad H_1 : \mu > 1.$$

This is a **one-sample test on the household changes**. Each household contributes one change, calculated from its before and after usage. The test is calculated using the boundary value $\mu = 1$.

(a-3) A colleague suggests checking whether the usage changes in (a-1) follow a normal distribution before conducting the hypothesis test. Is it necessary for these individual changes to be normally distributed? Explain your answer. (2 points)

No. The **Central Limit Theorem** tells us that with a large sample of independent observations, the sample mean can have an approximately normal distribution even when the individual observations do not. Here, we have 2012 households, so we can use this approximation to test the population mean. We do not need the household usage changes themselves to follow a normal distribution.

(a-4) Conduct the hypothesis test at a 5% significance level, using a 95% confidence level in R. Report the sample mean, test statistic, and p-value. Is there enough evidence that the mean increase for this group exceeds 1 hour per week? (3 points)

```
n <- length(hours_change)
mean_change <- mean(hours_change)
se <- sd(hours_change) / sqrt(n)
cat("Sample mean:", round(mean_change, 4), "hours\n")
```

```
## Sample mean: 1.476 hours
```

```
cat("Sample mean:", round(mean_change * 60, 2), "minutes\n")
```

```
## Sample mean: 88.56 minutes
```

```
cat("Standard error:", round(se, 4), "hours\n")
```

```
## Standard error: 0.2782 hours
```

```
# One-sample, one-tailed test of the mean change against 0.5 hours.
test_result <- t.test(hours_change, mu=1, alternative="greater",
                      conf.level=0.95)
test_result
```

```
##
##  One Sample t-test
##
## data:  hours_change
## t = 1.7111, df = 2011, p-value = 0.04361
## alternative hypothesis: true mean is greater than 1
## 95 percent confidence interval:
##  1.018211      Inf
## sample estimates:
## mean of x
##  1.475984
```

The sample mean increase is 1.4760 hours, or approximately **88.56 minutes per week**. The test statistic is 1.711, and the p-value is approximately **0.0436106**. Since the p-value is below 0.05, we **reject the null hypothesis**. There is statistically significant evidence that the population mean increase exceeds 60 minutes per week among households subscribed both before and after the launch. The one-sided 95% confidence interval has a lower bound of approximately 61.09 minutes, which is also above 60 minutes.

(a-5) Does a statistically significant result in (a-4) show that the gaming features caused a great-than-60-minute-per-week increase in Netflix usage? Explain. (2 points)

No. The statistically significant result provides evidence that average usage increased by more than 60 minutes in this group, but it does not show that the gaming features caused that increase. For example, Netflix may have released popular new shows during the same

period, which could increase usage even without gaming features. A before-and-after comparison alone cannot tell us how usage would have changed if the gaming features had not been introduced. The result also applies only to households subscribed in both periods; it does not describe households that joined or canceled Netflix.

(b) Competition with Disney+ and Hulu

(b-1) Using the data from both before and after the beta launch, calculate two averages of combined weekly Hulu and Disney+ usage: one for Netflix subscribers and one for non-subscribers. Which group has the higher average usage of Hulu and Disney+, the group with Netflix subscription or the group without? (2 points)

```
# Reload the full dataset, including households excluded from part (a).
streaming_beta <- read.csv("streaming_beta.csv")
# Combined weekly Hulu+Disney hours in each period.
streaming_beta$hd_pre <- streaming_beta$hulu_hours_pre +
                         streaming_beta$disney_hours_pre
streaming_beta$hd_post <- streaming_beta$hulu_hours_post +
                          streaming_beta$disney_hours_post
# Combine the observations from both periods.
both_periods <- data.frame(
  household_id = rep(streaming_beta$household_id, 2),
  has_netflix = c(streaming_beta$has_netflix_pre,
                  streaming_beta$has_netflix_post),
  hd_hours = c(streaming_beta$hd_pre, streaming_beta$hd_post)
)
subscriber_hours <- both_periods$hd_hours[both_periods$has_netflix == 1]
nonsubscriber_hours <- both_periods$hd_hours[both_periods$has_netflix == 0]
usage_summary <- data.frame(
  Group = c("Netflix subscribers", "Netflix non-subscribers"),
  Observations = c(length(subscriber_hours), length(nonsubscriber_hours)),
  Mean_hours = c(mean(subscriber_hours), mean(nonsubscriber_hours))
)
usage_summary
```

```
##                     Group Observations Mean_hours
## 1     Netflix subscribers         4698   4.015937
## 2 Netflix non-subscribers         3302   3.781280
```

```
mean_difference <- mean(subscriber_hours) - mean(nonsubscriber_hours)
cat("Difference in sample means:", round(mean_difference, 3),
    "hours per week\n")
```

```
## Difference in sample means: 0.235 hours per week
```

The average combined usage is approximately **4.016 hours** among Netflix subscribers and

Table 3: Average Weekly Hulu and Disney+ Usage Using Both Periods

Netflix Status	Number of Observations	Mean Hours
Subscriber	4698	4.016
Non-subscriber	3302	3.781

3.781 hours among non-subscribers. Netflix subscribers have a higher sample mean by approximately **0.235 hours**, or 14.1 minutes per week. The table counts observations, not distinct households: each household appears twice. A switcher contributes one observation to each group, while a household whose subscription status did not change contributes both observations to the same group.

(b-2) Using the findings in (b-1), can you conclude that Netflix does not compete with Disney+ and Hulu? Explain. (1 point)

No. Households that enjoy streaming more may subscribe to several platforms and spend more time on all of them. This can explain why Netflix subscribers have higher Hulu and Disney+ usage in the sample, even if Netflix takes away some of their viewing time from those platforms. Comparing different households does not tell us how much the same household would use Hulu and Disney+ if it did not subscribe to Netflix.

(b-3) Now let's focus on **switchers**: households that joined or canceled Netflix between the two periods. For each switcher, calculate its combined Hulu and Disney+ usage **with Netflix minus without Netflix**. State the null and alternative hypotheses, then use a one-sample t-test to test *whether the population mean of these differences is zero*, at a 5% significance level. Report the number of switchers, the mean difference, the p-value, and your conclusion.

Let D be a switcher's combined Hulu+Disney hours in the period with Netflix minus its combined hours in the period without Netflix. Let μ_D be the population mean of these differences among switchers. The hypotheses are:

$$H_0 : \mu_D = 0 \qquad \text{versus} \qquad H_1 : \mu_D \neq 0.$$

Each household provides **one difference**. We use a two-tailed test because the question asks whether the population mean difference is different from zero in either direction. A negative difference means that the household used Hulu and Disney+ less during the period with Netflix. Assume the differences are independent across households.

```
# Switchers subscribed in exactly one of the two periods.
switchers <- streaming_beta[
  streaming_beta$has_netflix_pre != streaming_beta$has_netflix_post,
]
# Identify the with-Netflix and without-Netflix periods for each household.
switchers$hd_with_netflix <- ifelse(
  switchers$has_netflix_pre == 1,
  switchers$hd_pre,
  switchers$hd_post
```

```
)
switchers$hd_without_netflix <- ifelse(
  switchers$has_netflix_pre == 1,
  switchers$hd_post,
  switchers$hd_pre
)
# Use the same direction for both joiners and cancelers.
switchers$difference <- switchers$hd_with_netflix -
                        switchers$hd_without_netflix
cat("Number of switchers:", nrow(switchers), "\n")
```

```
## Number of switchers: 674
```

```
cat("Mean hours with Netflix:",
    round(mean(switchers$hd_with_netflix), 3), "\n")
```

```
## Mean hours with Netflix: 3.301
```

```
cat("Mean hours without Netflix:",
    round(mean(switchers$hd_without_netflix), 3), "\n")
```

```
## Mean hours without Netflix: 5.059
```

```
cat("Mean difference (with minus without):",
    round(mean(switchers$difference), 3), "hours\n")
```

```
## Mean difference (with minus without): -1.757 hours
```

```
switcher_test <- t.test(switchers$difference, mu=0,
                        alternative="two.sided", conf.level=0.95)
switcher_test
```

```
##
##  One Sample t-test
##
## data:  switchers$difference
## t = -5.6665, df = 673, p-value = 2.162e-08
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  -2.366250 -1.148387
## sample estimates:
## mean of x
## -1.757319
```

There are **674 switchers**. Their average combined usage is 3.301 hours with Netflix and 5.059 hours without Netflix. The mean difference is **-1.757 hours**, or about 105.4 fewer minutes per week with Netflix. The test statistic is -5.666 and the p-value is 2.16e-08, which is much smaller than 0.05. We **reject the null hypothesis**: among switchers, mean

Hulu+Disney usage is significantly lower in the period with Netflix.

(b-4) Why might the comparison across all households in (b-1) and the comparison among only switchers in (b-3) give different results? Explain what the results suggest about Netflix's competition with Disney+ and Hulu. (2 bonus points)

Part (b-1) compares different households, which may have different levels of interest in streaming. Part (b-3) compares the same household during its periods with and without Netflix. By doing so, we can investigate whether the usage of Hulu and Disney+ comes at the expense of Netflix—when these households gain/lose access to Netflix, are they going to decrease/increase their usage of Hulu and Disney+? Such relationship is more likely to provide causal evidence on the competition between Hulu/Disney+ and Netflix.